SolidJS

Is SolidJS actually faster and better than React?

Origin of interest

A *tweet* from PocketJS

> **React/React Native** is too slow on handheld consoles, etc.

> **SolidJS** represents new performance baseline of Javascript UI.

- PocketJS

Tweet



Sources:
https://x.com/pocket_js/status/2094061588735205471
https://pocketjs.dev/docs/overview/

What I was using is slow?

React and React Native

React

> **Library** that let us create components.

> Next.js is **recommended** for full-stack apps.

> *"People expect web app pages to **load fast**"*

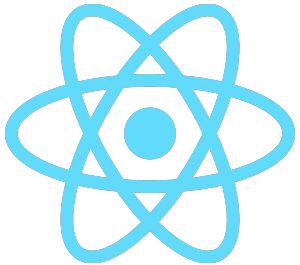
Target: Browsers

React Native

> Software **framework** for native development.

> *"Learn once, write anywhere"*

Target: Android, iOS



Sources:
https://react.dev/
https://reactnative.dev/

Was my curiosity still valid?

Basic benchmark app

Goals:

> Creating the same app in both **SolidJS** and **React**.

> Design E2E tests with **Playwright**.

> Compare performance between both instances (is faster).

> Compare subjectively both experiences (is better).

What is SolidJS?

SolidJS

> JavaScript **framework** designed to build high-performing user interfaces.

> Reactivity.

> **Fine-grained** updates.



Sources:
https://docs.solidjs.com/

SolidJS: Fine-grained reactivity

> Connection of UI elements with data.

> Ability of make highly **targeted** and **specific** updates.

> **SolidJS:** *"updates are made to the **targeted attribute** that needs to be changed"*

> **React:** *"would re-execute an **entire component** for a change in the single attribute"*



Sources:
https://docs.solidjs.com/advanced-concepts/fine-grained-reactivity

What is Playwright?

Playwright

> **Framework** for web automation and testing.

> **Test runner** built for end-to-end testing (Playwright Test).

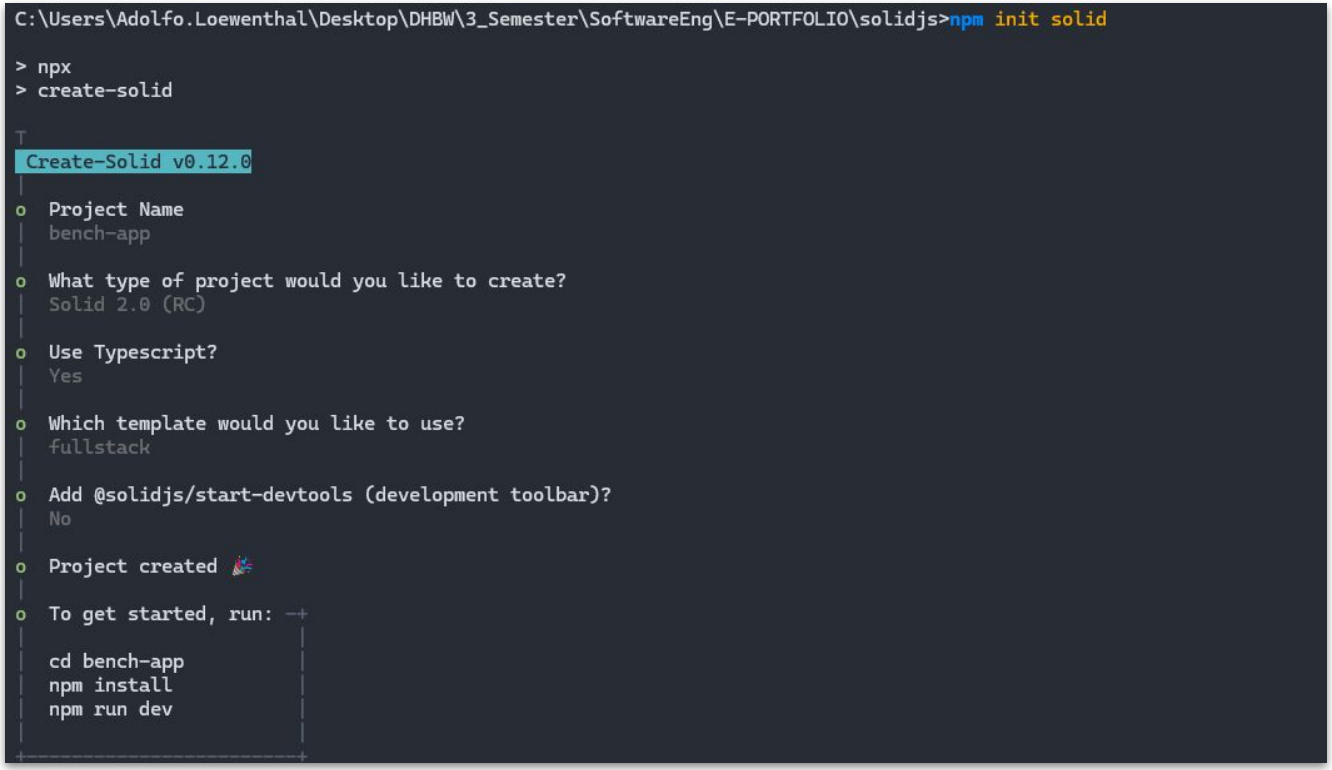
> Runs tests across Chromium, Firefox, and WebKit.



Sources:
https://github.com/microsoft/playwright

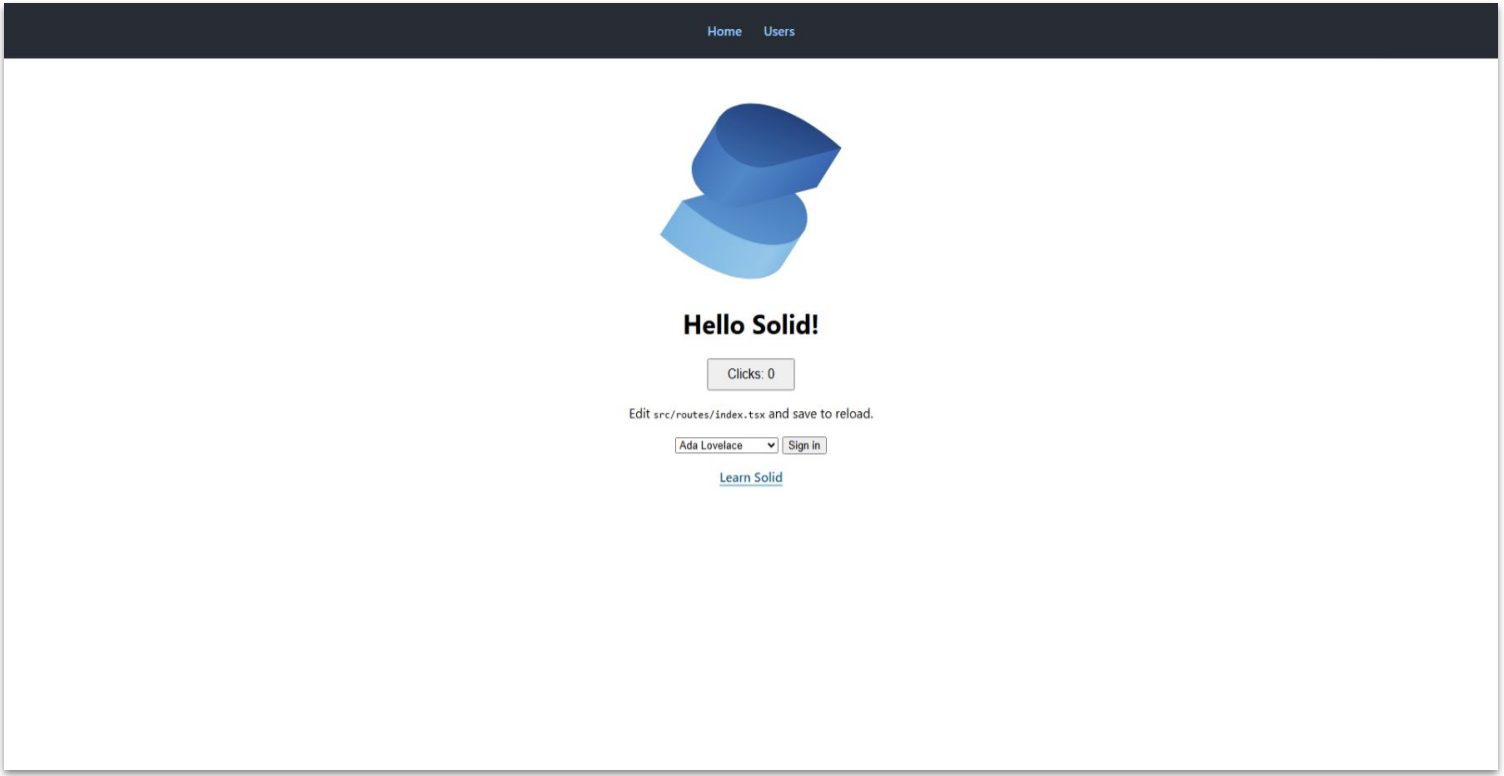
Setting up experience

SolidJS 2.0 (RC) + Vite: npm init solid

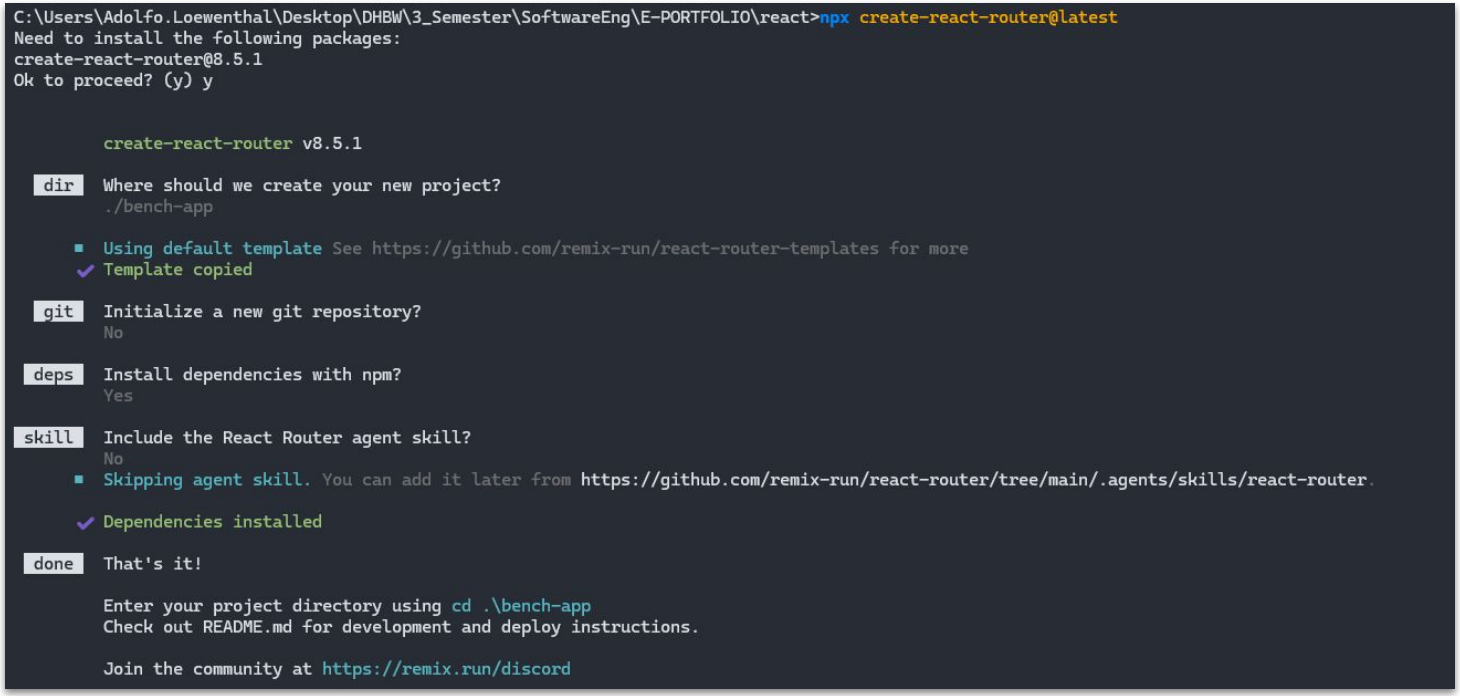


Sources:
https://docs.solidjs.com/quick-start

SolidJS 2.0 (RC) + Vite: first page

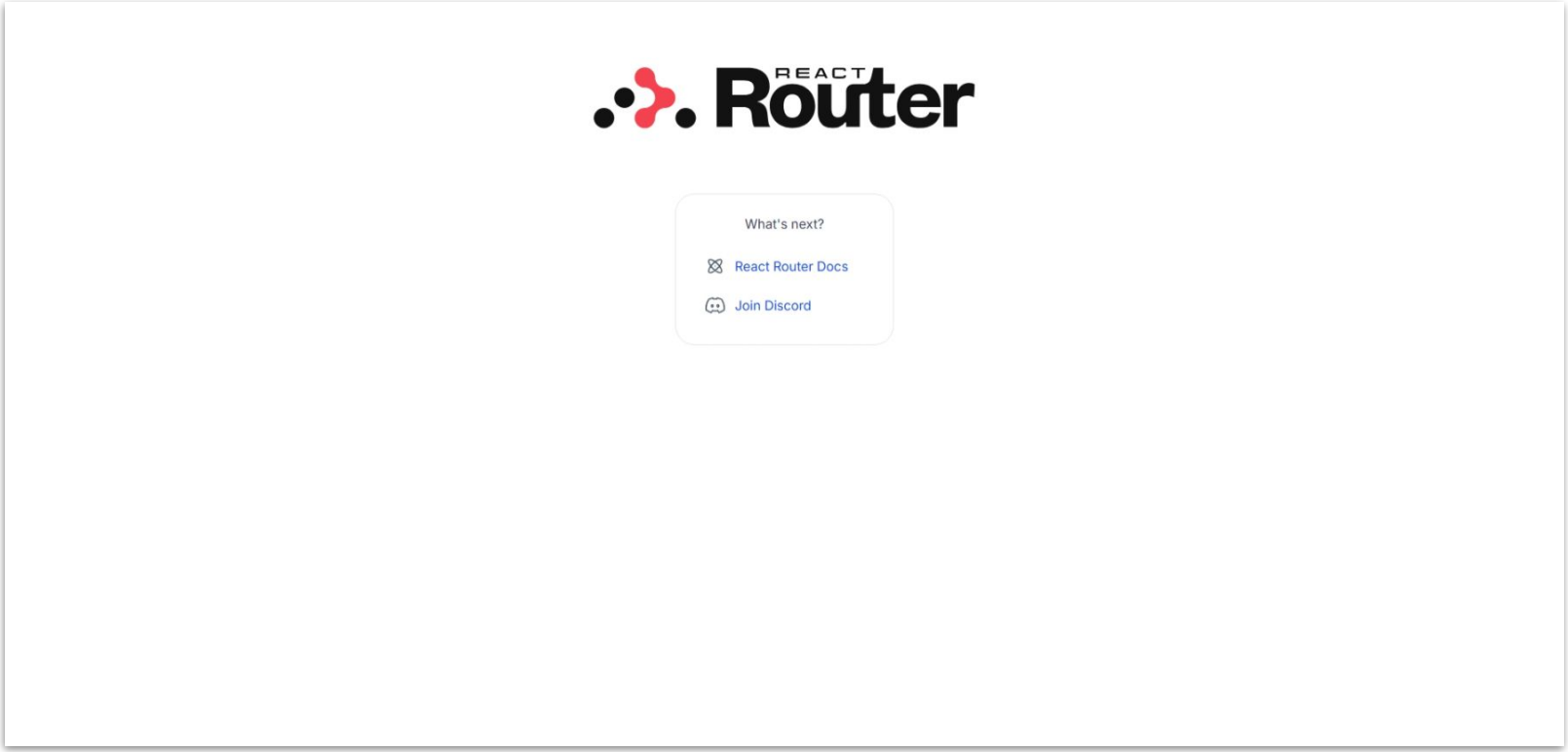


React Router + Vite: npx create-react-router



Sources:
https://react.dev/learn/creating-a-react-app#react-router-v7

React Router + Vite: first page



Code comparison

SolidJS

```tsx
import { createSignal, createStore, For } from 'solid-js';
import { Title } from '@solidjs/meta';
import CardList from '../../components/CardList';
import Card from '../../components/Card';
import TrafficLight from '../../components/TrafficLight';
import { Profile } from '../../lib/profile';

export default function FormsIndex() {
  const [name, setName] = createSignal('');
  const [street, setStreet] = createSignal('');
  const [phone, setPhone] = createSignal('');
  const [count, setCount] = createSignal(0);
  const [traffic, setTraffic] = createSignal(0);

  const [profiles, setProfiles] = createStore<Profile[]>([]);

  const createProfile = () =>
    setProfiles((profile) => {
      profile.push({
        name: name(),
        street: street(),
        phone: phone()
    });
    setCount(count()+1);
    handleTraffic();
  });

  const handleTraffic = () => {
    setTraffic(traffic()+1);
    if(traffic() == 2)
      setTraffic(0);
  };

  return (
    <>
    </>
  );
}
```

React

```tsx
import { useState } from "react";
import Card from "~/components/Card";
import CardList from "~/components/CardList";
import TrafficLight from "~/components/TrafficLight";
import type { Profile } from "~/lib/profile";

export default function FormPage() {
  const [name, setName] = useState('');
  const [street, setStreet] = useState('');
  const [phone, setPhone] = useState('');
  const [count, setCount] = useState(0);
  const [traffic, setTraffic] = useState(0);

  const [profiles, setProfiles] = useState<Profile[]>([]);

  const createProfile = () => {
    setProfiles([
      ...profiles, {
      name: name,
      street: street,
      phone: phone
    }]);
    setCount(count+1);
    handleTraffic();
  };

  const handleTraffic = () => {
    setTraffic(traffic+1);
    if(traffic == 2)
      setTraffic(0);
  };

  return (
    <>
    </>
  );
}
```

Pros and cons

SolidJS

Pros

> Code is **easy to migrate** from and to React.

> Easy interactive installation.

> **Fine-grained** reactivity.

Cons

> In transition to **Solid 2.0 RC** (13.08.26), so docs are still raw.

> It's still a **niche** framework.

Sources:
https://www.solidjs.com/blog/solid-2-0-rc-the-big-reveal
https://www.toptal.com/developers/react/solidjs-vs-react

React

Pros

> Code is **easy to migrate** from and to SolidJS.

> Easy interactive installation.

> **Top 2** most popular web frameworks globally (+40%).

Cons

> Re-renders components by **default**.

> **Virtual DOM + Real DOM** to compare changes.

Sources:
https://www.statista.com/statistics/1124699/worldwide-developer-survey-most-used-frameworks-web
https://www.toptal.com/developers/react/solidjs-vs-react

Live demo